EPOCH bugs and improvements

Contributors to this register

Official `main` checked 30 October 2026

1 How to read this document

This is a shared list of bugs and small improvements to EPOCH. The official `main` branch is the reference point. For bugs, the colour shows where a fix is available: **Fix: main** in official `main`, **Fix: 4.20-devel** in an official development branch, **Fix: fork** in a separate copy of EPOCH, or **Not fixed** if no fix is known. The yellow flag can name any official development branch. A purple flag does not mean the fix has reached official EPOCH. For improvements, the colour shows where the improvement is available: **Main**, **4.20-devel**, **Fork**, or **Not available**. The branch checked for this edition of the register is `f294c484f`. The tables are short; click an ID for the explanation and source links. These checks compare EPOCH files and commit records. They are not new EPOCH runs.

2 Bugs

Bug list

Bug IDs are numbers. Keep each description short; the middle column wraps when needed. Long paths and test results belong below the table.

ID	Bug	Fix status
1	Laser absorption diagnostic at maximum boundaries	**Fix: main**
2	2D linear Breit–Wheeler selects the wrong species	**Fix: main**
3	Photon probes can deadlock with particle IDs	**Fix: 4.20-devel**
4	Particle IDs can be reused after restart or file loading	**Fix: fork**
5	Higuera–Cary particle push uses the wrong field scale	**Fix: fork**
6	Trident can produce pairs when pair production is disabled	**Fix: fork**
7	Pair creation can use the wrong photon species	**Fix: fork**
8	`identify` can silently replace charge or mass	**Fix: fork**
9	Spin state can be lost on restart in the fork	**Fix: fork**
10	Boosted pair channels have sampling and scaling errors in the fork	**Fix: fork**
11	Custom-laser reader samples half a cell away in the fork	**Fix: fork**
12	Custom-laser file loading can deadlock across MPI ranks in the fork	**Fix: fork**
13	Two custom lasers can share the wrong file data in the fork	**Fix: fork**
14	3D custom-laser profile can become stale after load balancing	**Fix: fork**

1. Laser absorption at maximum boundaries

Energy injected at `x_max`, `y_max` or `z_max` could be recorded with the wrong sign. This could make the reported absorption coefficient zero. The fix covers 1D, 2D and 3D laser code. Tom Goffrey merged PR #896 on 10 April 2026 as `0984b1d31`. It is in official `main` and in the v4.20.1 release notes (issue #893).

2. Linear Breit–Wheeler species selection in 2D

A one-line typo in the 2D linear Breit–Wheeler routine selected the wrong species. PR #892 merged `ba04ae645` on 26 March 2026. The fix is in official `main` and listed in v4.20.1 (issue #891). This is separate from the nonlinear Breit–Wheeler items below.

3. Photon-probe particle-ID deadlock

When it happens

With photon probes and `PARTICLE_ID` or `PARTICLE_ID4`, a rank with no photons could return before a call that all ranks needed to make. Other ranks would wait indefinitely.

Where the fix is

PR #906 merged `241d6fb9c` into the official `4.20-devel` branch on 1 June 2026, in all three dimensions. It is not in the official `main` version checked for this document. A later 2D fork commit, `3e4734522`, also changes this path but includes a separate ID fix (bug 4).

Test to keep

Run with at least two MPI ranks, a photon probe, and one rank without photons. Check that the run finishes and that the IDs remain unique. One rank cannot expose this deadlock.

4. Particle IDs reused after restart or file loading

Problem. The 2D investigation found that the largest assigned ID was not always carried through restart or particle loading from a file. New particles could then reuse an old ID.

Fix and status. `3e4734522` saves and raises the ID counter in the fork. The checked official `main` does not contain those changes. The fork fix is 2D only. A regression test should check IDs after particles leave the domain, after a restart with a different MPI layout, and after loading both supplied and automatic file IDs.

5. Higuera–Cary particle-push field scale

Problem

The magnetic rotation in the Higuera–Cary particle push uses an interpolated field. Its scale must include the same factor used elsewhere in the push. In the checked official `main`, that factor is missing from this rotation.

Fix and test

The fork fix `d86ceceab` changes the calculation in 1D, 2D and 3D. A separate branch based on official `4.21-devel` contains `2684956bf`. Neither change is in official `main`. A useful test is a single particle in crossed fields, compared with an independent trajectory, for several particle shapes. This document checks the source; it does not rerun that test on the official build.

6. Trident pair creation with pairs disabled

Problem

With `TRIDENT_PHOTONS` compiled and `produce_pairs = F` in the deck, the electron loop could still try to create a trident pair. If no pair species was set, this could use species index `-1` and crash.

Fix and test

The fork commit `1a88d4afa` checks `produce_pairs` before changing the trident event counter and before creating a pair, in all three dimensions. The checked official `main` still has the unguarded path. A test should run past a would-be event with pairs disabled and confirm that the run completes without creating a pair.

7. Wrong photon species in nonlinear pair creation

The 2D photon loop in the checked official `main` passes `photon_species` to pair creation instead of the species currently being processed, `ispecies`. With more than one photon species, this can use the wrong particle list and may crash. The 2D fork fix `2b18ef6b4` passes `ispecies`. A test needs at least two photon species that can create nonlinear Breit–Wheeler pairs.

8. `identify` silently replaces charge or mass

In an EPOCH species block, `identify` can replace an explicit `charge` or `mass` without warning. A deck may then run with different particle properties than its author expected. The fork commit `b908c8591` adds a warning in all three dimensions. It does not change which properties EPOCH uses. The warning is absent from the checked official `main`. Tests should cover both input orders and both matching and mismatching values.

9. Spin state lost on restart in the fork

This affects the fork's 2D spin feature; the feature is not in the checked official `main`. The fork fix `a66b394cb` restores stored spin state on restart. A test should compare a continuous run with a restarted run.

10. Boosted pair-channel errors in the fork

This entry groups related fixes to pair-boost features absent from the checked official `main`. They are separate code changes: Breit–Wheeler boost handling (`3e9f3121f`), muon gamma sampling (`fbb9a9459`), Bethe–Heitler scaling (`92552de69`, `a5716caf2`), and additive event counters (`99ce3c685`). Review and test each change separately before proposing it for official EPOCH.

11. Half-cell offset in the fork's custom-laser reader

The fork's 2D file-based laser reader could sample a profile half a cell away from the intended point. The correction is `1b307b37d`. The checked official `main` does not have this reader.

12. MPI deadlock in the fork's custom-laser reader

Problem and fix

Ranks could take different paths while loading profile and phase files, leaving some ranks waiting. The 2D fork fix is `5a77c367b`. This concerns a reader absent from the checked official `main`.

Test to keep

Load both profile and phase files on multiple ranks, including a rank with no requested local data. All ranks should finish and the injected field should match the supplied data.

13. Two custom lasers sharing file data in the fork

The 2D fork reader could let one laser block use another block's imported profile. `945fd060d` stores the data with each laser block. The reader is absent from the checked official `main`. A test should load two different profiles on the same boundary and check both fields.

14. Stale 3D custom-laser profile after load balancing

In the fork's 3D reader, the loaded profile window needed to follow changes to the domain split. `f572d035c` reloads the relevant profile portion as the split moves. The reader is absent from the checked official `main`. A test should compare the injected field before and after a load-balancing change.

3 Improvements

Improvement list

These entries describe changes to a method or feature. The last column shows where each improvement is available; it does not claim that a missing improvement is a bug.

ID	Improvement	Available in
I01	Numerical nonlinear Breit–Wheeler rate table	**Fork**
I02	Polarisation-resolved pair energy tables	**Fork**
I03	More accurate spin-resolved photon energy sampling	**Fork**
I04	Sub-step spin and polarisation update	**Fork**
I05	Full Landau–Lifshitz force and midpoint radiation-reaction step	**Fork**

I01. Numerical nonlinear Breit–Wheeler rate table

Nonlinear Breit–Wheeler (NBW) pair creation is the conversion of a high-energy photon into an electron–positron pair in a strong electromagnetic field. Official 2D EPOCH uses an approximate analytical rate in `pairprod.table`. The 2D fork adds a numerical table for the locally constant field approximation (LCFA), `pairprod_lcfa.table` (`848d21a4b`), selected through `pair_tables = lcfa` (`fd56a2e08`). The old method remains available as `legacy`. This is an improvement candidate, not a claim that the old method is a coding error. It can change simulation results, so record the chosen table and its hash. The local table generator is currently not committed to the fork; clean regeneration should be documented before proposing this for official EPOCH.

I02. Polarisation-resolved pair energies

The fork's 2D spin feature adds two tables for the energy shared between a pair, depending on photon polarisation (`dfb93b141`, `2b18ef6b4`). The checked official `main` does not contain that spin feature. With spin enabled in the fork, `pair_tables = legacy` restores only the old *rate*; the polarisation-resolved energy split remains.

I03. Spin-resolved photon-energy sampling

The fork adds tables that sample nonlinear Compton photon energies more accurately when spin is tracked in 2D (`6610b3768`). The switch is `use_nlcs_quantile_tables`. Runs without the fork's spin feature continue to use EPOCH's older sampling table. This change is absent from the checked official `main`.

I04. Sub-step spin and polarisation update

The fork's 2D spin model updates spin and photon polarisation within a simulation time step, including steps without an emission event (`944fff881`). This is not a change to every EPOCH emission mode, and it is absent from the checked official `main`.

I05. Fuller radiation-reaction model and time step

The 2D fork adds optional terms of the full Landau–Lifshitz radiation-reaction force, another Gaunt-factor fit, and a midpoint time step for the radiation-reaction force (`dfa48930a`). The old options remain the defaults. These additions are not in the checked official `main`; the audited fork branch has no equivalent 3D version.

4 Adding an entry

Add one short row to `registry.tex` or `improvements_registry.tex`, then add a matching entry to `items.tex` or `improvements.tex`. Use a subsection for a serious bug and smaller headings for the failure, fix and test. Record the official branch checked and the date when changing a flag. The copyable examples are in `CONTRIBUTING.md`.